

Sai Krishna Mateti

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EDUCATION

- University of Alabama at Birmingham, Alabama** Aug 2023 - May 2025
Master of Science in Electrical and Computer Engineering GPA: 3.91 / 4.0
Relevant Coursework: Machine learning, Data analysis, Computer vision, Digital image processing

SKILLS SUMMARY

- Languages:** Python, Java, JavaScript, TypeScript, HTML, CSS
- Frameworks/Libraries:** React, Node.js, TensorFlow, LangChain, OpenCV
- Tools & Platforms:** Git, GitHub, VS Code, IntelliJ, Jupyter, Google Colab, Linux, Windows, Arduino, Raspberry Pi
- Concepts:** Data Structures, REST APIs, Version Control, Agile Methodologies, Machine Learning

EXPERIENCE

- BearCreek Ai** Birmingham, US
Software Engineer Oct 2024 - Currently Working
 - Developed and deployed a real-time AI chat interface used by **1,500+ users** across educational institutions.
 - Integrated **LLM agents (OpenAI o1, DeepSeek r1)** with **LangChain** to support Responsive AI interactions and multi-model compatibility, reducing user query latency by **~30%**.
 - Optimized full-stack architecture and API request handling to scale for **1,000+** concurrent sessions, maintaining **99.9%** uptime and fast response times.
 - Collaborated with a 4-member agile team using **GitHub version control**, completing sprint cycles with an average code merge time of under **24 hours**.
 - Delivered production-ready features with CI/CD best practices, contributing to system-wide **~50% reduction in bug reports** over three release cycles.
- EquipSense** Birmingham, US
Software Engineer Intern (Internship) July 2024 - Sep 2024
 - Built a responsive cloud dashboard for **real-time asset tracking** used by **3 pilot clients** in logistics.
 - Integrated GPS tracking with IoT data to improve visibility across **100+ deployed assets**.
- UAB Recreational Center** UAB Campus
Supervisor / Facility Operations Staff (Part-time) Feb 2024 - July 2024
 - Supervised daily operations of a **125,000 sq. ft.** facility serving **1,000–1,500 students daily**, managed **3+ staff**, enforced university policies, and ensured high-quality service and safety compliance.

TECHNICAL / PERSONAL PROJECTS

Real-Time Vehicle & Person Detection with YOLOv11 Segmentation(link) | Python Jan. 25 - April. 25

- Developed YOLOv11 pipeline with **91.2% mAP@0.5** and **78.9% IoU** for real-time traffic segmentation.
- Detected **8,000+ objects** at 30 FPS in 720p video using PyTorch and OpenCV.
- Boosted post-processing speed by **20%** using Supervision for segmentation overlays.

Analyzing Expenditures for Insurance Company Customers(link) | Machine learning May. 2023 - July 2023

- Analyzed a dataset of **1,300+ insurance customers** and built a linear regression model to predict medical expenses with a model accuracy of **92%**.
- Implemented **gradient descent** from scratch to optimize a regression model, improving training efficiency and accuracy.
- Forecasted future insurance costs using trained model on test data, supporting potential **cost planning improvements** of up to **25%** for underwriting processes.

Home Security System Using Arduino With IOT (link) | C/C++ February. 2022 - May. 2022

- Developed an IoT-based home security system using **Arduino** and **sensor modules**, enabling remote monitoring via smartphones and PCs for **24/7 access**.
- Integrated centralized data collection from **4+ sensors** enhancing real-time detection of security breaches.

EXTRA-CURRICULAR ACTIVITIES

- Technical coordinator:** Organized events for 2,000+ attendees and managing 10+ student volunteers. 2019-2021
- Volunteer Experience:** Contributed a valuable services for **(ICIEEE-2018)**. 2018